

Computer Networking: Security

(CSE 3752)

Experiment 6

Aim:

To implement asymmetric key cryptography using RSA algorithm and perform Diffie–Hellman key exchange to establish a shared secret key between communicating parties.

Objectives:

1. Understand the working of the RSA public key cryptosystem and the concept of Diffie–Hellman key exchange.
2. Perform encryption and decryption using RSA.
3. Compute shared secret key using Diffie–Hellman algorithm.
4. Analyze how asymmetric cryptography supports secure communication.

Exercises:

1. Given the following values for the RSA algorithm:

- Two prime numbers: $p=61$, $q=53$
- Public key: $e=17$

Calculate the private key d , where d is the modular inverse of e modulo $\phi(n)$.

2. Given the following parameters for Diffie-Hellman algorithm used by A and B for key exchange.

- The shared prime $q=157$ and the primitive root $p=5$.

Calculate: (a) The value of Y_A and Y_B transmitted by both A and B.

(b) The value of secured key (K) shared by both A and B.

3. Given a scenario, where B has received a document from A through internet. Explain, how B confirms that the document has been transmitted by A only (not any adversary) using the concept of digital signature as an use of public key cryptosystem.
4. Transform the plaintext "AES USES A MATRIX" into a state matrix form.
5. Compute the output of S-box with given input state matrix as $\begin{bmatrix} 4 & 5 \\ E & 2 \end{bmatrix}$ and S-box as in Fig. 1.

	00	01	10	11
00	1001	0100	1010	1011
01	1101	0001	1000	0101
10	0110	0010	0000	0011
11	1100	1110	1111	0111

Fig. 1.

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$$

Fig. 2.

6. Perform ShiftRow transformation on the given current matrix in Fig. 2.
7. Find the output of Mix Column if the input state matrix is $\begin{bmatrix} D & 1 \\ A & F \end{bmatrix}$ with the predefined matrix as $\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$ and the irreducible polynomial for $GF(16)$ is x^4+x+1 .

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OBJECTIVES

1. Understand the working of the RSA public key cryptosystem and the concept of Diffie-Hellman key exchange.
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4. Analyze how asymmetric cryptography supports secure communication.

THEORY

① Understanding the working of RSA public key cryptosystem and Diffie-Hellman key exchange:-

- RSA Algorithm is an asymmetric encryption algorithm that uses two keys:
 - Public key for encryption
 - Private key for decryption.

It uses two prime numbers to generate keys.

Encryption and decryption are performed using modular arithmetic.

Encryption: $C = M^e \text{ mod } n$.

Decryption: $M = C^d \text{ mod } n$

Diffie-Hellman Key Exchange is used to securely generate a shared secret key between two users over an insecure network.

Both users generate public keys and compute the same secret key independently.

② Analyze how asymmetric cryptography supports secure communication:-

- Asymmetric Cryptography uses a pair of keys:

- Public key
- Private key

It supports secure communication by:

- Providing confidentiality
- Ensuring authentication
- Maintaining data integrity.
- Enabling security key exchange
- Supporting digital signatures

It is widely used in secure communication systems and network security

CODE

Objective 2

#RSA .Encryption and Decryption.

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a  
  
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:  
            return d  
  
def encrypt(msg, e, n):  
    return [pow(ord(ch), e, n) for ch in msg]  
  
def decrypt(cipher, d, n):  
    return ''.join(chr(pow(c, d, n)) for c in cipher)  
  
p = int(input("Enter p: "))  
q = int(input("Enter q: "))  
e = int(input("Enter e: "))  
msg = input("Enter message: ")  
  
n = p * q  
phi = (p - 1) * (q - 1)  
  
if gcd(e, phi) != 1:  
    print("e and phi are not coprime")  
    exit()  
  
d = mod_inverse(e, phi)  
  
print("\nn = ", n)  
print("phi(n) = ", phi)  
print("Public key = ", (e, n))  
print("Private key = ", (d, n))  
  
cipher = encrypt(msg, e, n)  
print("Ciphertext = ", cipher)  
  
plain = decrypt(cipher, d, n)  
print("Decrypted Message = ", plain)
```


OUTPUT

Enter p: 61
 Enter q: 53
 Enter e: 17
 Enter message: HELLO
 $n = 3233$
 $\phi(n) = 3120$
 Public key = (17, 3233)
 Private key = (2753, 3233)
 Ciphertext = [3000, 28, 2726, 2726, 1307]
 Decrypted Message = HELLO

• Objective 3

Diffie - Hellman Key Exchange

```

g = int(input("Enter prime number g: "))
g = int(input("Enter primitive root g: "))
xA = int(input("Enter private key of A: "))
xB = int(input("Enter private key of B: "))
yA = pow(g, xA, g)
yB = pow(g, xB, g)
print("\n Public key of A = ", yA)
print("Public key of B = ", yB)
KA = pow(yB, xA, g)
KB = pow(yA, xB, g)
print("\n Secret key for A = ", KA)
print("Secret key for B = ", KB)
if KA == KB:
    print("Shared Secret key = ", KA)
else:
    print("Key Exchange Failed")
  
```

OUTPUT:

Enter prime number g: 157
 Enter primitive root g: 5
 Enter private key of A: 15
 Enter private key of B: 27
 Public key of A = 79
 Public key of B = 65
 Secret key for A = 78
 Secret key for B = 78
~~Secret key for A = 78~~
 Shared Secret key = 78

ANALYSIS

① Perform encryption and decryption using RSA:-

- Two prime numbers p and q are used to generate RSA keys.
- The modulus n and Euler's Totient $\phi(n)$ are calculated.
- Public key (e, n) and private key (d, n) are generated.
- Plaintext is converted into encrypted ciphertext using the public key.
- Ciphertext is decrypted using the private key.
- The original plaintext is recovered successfully after decryption.
- RSA provides secure ~~communication~~ communication using asymmetric encryption.

② Computing shared secret key using Diffie-Hellman algorithm:-

- Shared prime number and primitive root were selected successfully.
- Both users generated their own private keys.
- Public keys were computed using modular exponentiation.
- Shared secret ~~key~~ keys were generated independently by both users.
- Both computed secret keys matched successfully.
- Secure key exchange was achieved over an insecure network.
- Diffie-Hellman enables secure communication without directly sharing the secret key.

CONCLUSION

The RSA Algorithm and Diffie-Hellman Key Exchange were successfully implemented for secure communication. RSA was performed by generating public and private keys, and the plaintext was encrypted and decrypted successfully using modular exponentiation. Diffie-Hellman key exchange was also implemented successfully to generate a common shared key between two users, demonstrating the role of asymmetric cryptography in secure communication over insecure networks.

EXERCISES

- Given the following values for the RSA algorithm:
 - Two prime numbers: $p=61, q=53$
 - Public Key: $e=17$
 Calculate the private key d , where d is the modular inverse of e modulo $\phi(n)$.
- Given the following parameters for Diffie-Hellman algorithm used by A and B for key exchange.
 - The shared prime $q=157$ and primitive root $p=5$.
 Calculate: (a) The value of x_A and y_A transmitted by both A and B.
(b) The value of shared key (K) shared by both A and B.
- Given a scenario, where B has received a document from A through internet. Explain, how B confirms that the document has been transmitted by A only (not any adversary) using the concept of digital signature as an use of public key cryptosystem.
- Transform the plaintext "AES USES A MATRIX" into a state matrix form.
- Compute the output of S-box with given input state matrix as $\begin{bmatrix} 4 & 5 \\ E & 2 \end{bmatrix}$ and S-box as in Fig 1.

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Fig 1

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Fig 2.

- Perform ShiftRow transformation on the given current matrix in Fig 2.
- Find the output of Mix Column if the input state matrix is $\begin{bmatrix} D & 1 \\ A & F \end{bmatrix}$ with the predefined matrix as $\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$ and the irreducible polynomial for $GF(16)$ is x^4+x+1 .

(1) Given,

- $p=61$
- $q=53$
- $e=17$

Calculation:

$$n = 61 \times 53 = 3233$$

$$\phi(n) = (61-1)(53-1) = 60 \times 52 = 3120$$

Now,

finding d such that,

$$17d \equiv 1 \pmod{3120}$$

$$\therefore d = 2753$$

$$\therefore \text{Public Key} = (17, 3233)$$

$$\text{Private Key} = (2753, 3233)$$

- (2) Given,
- $q = 157$
 - $g = 5$
 - $X_A = 15$
 - $X_B = 27$

Public keys:-

$$Y_A = 5^{15} \text{ mod } 157 = 79$$

$$Y_B = 5^{27} \text{ mod } 157 = 65$$

Shared secret key

$$K = 65^{15} \text{ mod } 157 = 78$$

$$\therefore Y_A = 79$$

$$Y_B = 65$$

Shared secret key = 78

- (3)
- (1) A creates a digital signature using its private key
 - (2) A sends the document and signature to B.
 - (3) B decrypts the signature using A's public key.
 - (4) If the decrypted hash matches the document hash, B confirms.

B confirms:

- The document is sent by A.
- The document is not modified by any adversary.

(4) Plaintext: AES USES A MATRIX

ASCII HEX values:

41 45 53 20
55 53 45 53
20 41 20 40
41 54 52 49

State matrix:

41 45 53 20
55 53 45 53
20 41 20 40
41 54 52 49

(5) Input Matrix:-

4 5
E 2

using the given S-Box:-

- $S(4) = 0010$
- $S(5) = 0000$
- $S(E) = 1111$
- $S(2) = 1010$

Output matrix:-

0010 0000
1111 1010

(6) Given Matrix:

63 E8 9F 40
10 2F 93 92
AB 30 AF C7
20 1B 2B A2

After ShiftRow:-

63 E8 9F A0
2F 93 92 10
AF C7 AB 30
A2 20 1B 2B

(7) Input Matrix:-

D 1
A F

Predefined Matrix:

1 4
4 1

Irreducible Polynomial:-
 $x^4 + x + 1$

Output Matrix:-

1 6
0 B

(ans).